

Projection Ambiguity in Thimble HMC and Its Parametrization-Level Resolution

Chien-Yu Chou,^{1,*} Jun Nishimura,^{2,3,†} and Cheng-Tsung Wang^{2,3,‡}

¹*RIKEN Center for Interdisciplinary Theoretical and Mathematical
Sciences (iTHEMS), Wako, Saitama 351-0198, Japan*

²*KEK Theory Center, Tsukuba, Ibaraki 305-0801, Japan*

³*SOKENDAI (The Graduate University for Advanced
Studies), Tsukuba, Ibaraki 305-0801, Japan*

Abstract

We investigate the projection step in Hybrid Monte Carlo algorithms formulated on flowed or thimble-like manifolds, where each molecular-dynamics update requires solving nonlinear constraint equations to return trial configurations to the target manifold. In practice, this projection step can fail or become numerically unstable, and such behavior is often interpreted as a manifestation of geometric obstruction associated with singular structures in complexified configuration space.

We show that this interpretation is incomplete. Even when the geometric definition of the manifold is fixed, the projection problem admits multiple parametrized realizations, and the resulting numerical behavior depends on how the constraint equations are presented to the solver.

To make this dependence explicit, we introduce a parametrization-level reformulation of the projection problem, based on a backflow representation and a tangent–normal decomposition (BTN). Within this framework, the same projection problem is reformulated as a parametrized system on which the solver acts.

Using this formulation, we compare two implemented computational setups for the same local projection problem under identical geometric conditions and show that projection stability, failure burden, and bias diagnostics can differ substantially across setups. In particular, we find that projection behavior is governed by the interface through which the geometric information relevant to the same local projection problem becomes numerically accessible to the solver.

These results clarify the interpretation of projection failures and provide a framework for analyzing their origin without conflating geometric properties with parametrization- or solver-dependent effects.

I. INTRODUCTION

The sign problem remains one of the central obstacles to nonperturbative computation in quantum many-body systems and quantum field theory.[1, 2] In path-integral formulations with a non-positive or complex weight, the Boltzmann factor cannot be used directly as a probability measure for importance sampling, and practical calculations must instead rely on phase or sign reweighting.[1, 2] In severe regimes, the average sign becomes exponentially

* chien-yu.chou@riken.jp

† jnishi@post.kek.jp

‡ ctwang@post.kek.jp

small, strongly degrading the efficiency of naive reweighting-based Monte Carlo estimates.[3] This is not merely a technical inconvenience: in a general setting, the fermionic sign problem is NP-hard, suggesting that no generic polynomial-time solution should be expected.[1] The same obstruction appears across key nonperturbative frontiers, including finite-density QCD, strongly correlated electron systems, and real-time quantum field theory, where highly oscillatory phase factors obstruct straightforward Monte Carlo sampling.[2–4] This is the setting in which contour-deformation approaches become relevant.

One response is to reformulate the integration domain itself. This motivates complexifying the integration variables and deforming the original contour into a manifold embedded in complexified configuration space.[5, 6] For holomorphic actions and observables, the original contour can be deformed within the same integration cycle, so that the integral is reformulated on geometrically more suitable complex contours.[5]

Picard–Lefschetz theory provides a geometric framework for such deformations.[5, 7] In this framework, the original integration cycle is decomposed into an integer linear combination of Lefschetz thimbles, each associated with a critical point and fixed by intersection numbers.[5] Along each thimble, defined by downward flow of the real part of the action, the imaginary part remains constant, converting a highly oscillatory phase factor into a constant phase contribution.[6]

The same framework extends to oscillatory path integrals, including real-time ones, by reformulating them on deformed complex contours or thimbles.[5, 7] Contour deformation therefore provides a principled way to reduce phase oscillations associated with the imaginary part of the action without changing the value of the path integral.[5, 6, 8]

The remaining issue is practical implementation. The Lefschetz-thimble framework provides the geometric foundation, but practical algorithms are usually formulated either on a single thimble or on a finite-flow deformed manifold in the same contour-deformation class.[6, 8–10] Early work showed that lattice quantum field theories can be defined on a single thimble and sampled stochastically.[6] Later studies confirmed substantial phase-fluctuation reduction in concrete models, while also showing that a residual measure phase remains and must be handled by reweighting.[9]

These practical bottlenecks led to algorithmic efforts to sample flowed or thimble-like manifolds more efficiently and to control residual-phase and ergodicity costs more explicitly.[8, 9] In particular, Hybrid Monte Carlo (HMC) has been formulated on flowed

manifolds by parametrizing configurations with flow-based variables, enabling global updates while controlling the residual phase.[10] That parametrization introduces additional computational structure, most notably Jacobian determinants whose direct evaluation scales as $\mathcal{O}(V^3)$ with system size.[11]

To mitigate both residual sign severity and computational cost, finite-flow-time approaches have been developed.[8, 12] In flowed-manifold approaches, one must balance sign-problem reduction against increasingly difficult sampling: too little flow leaves strong phase fluctuations, whereas too much flow can trap the Markov chain behind action barriers.[8, 12] This pressure falls not only on the choice of flow time but also on the choice of parametrization.

Rather than varying flow time alone, one can also change how the flowed manifold is parametrized. In these flowed-manifold formulations, configurations are no longer updated in unconstrained Euclidean space but evolve on a curved manifold embedded in complexified field space.[8, 10] Operationally, this introduces a projection step: a molecular-dynamics update produces a trial configuration that must be returned to the target manifold before the trajectory can continue. The projection step is therefore a natural place to analyze practical numerical behavior.

Numerical difficulties in this step are often discussed in geometric terms, especially when the flowed manifold approaches singular regions such as determinant zeros or when the conventional gradient flow develops finite-time blow-up.[13] At the same time, the observed numerical behavior depends on how the geometric information relevant to the projection problem is carried into the parametrization layer and on how the constrained dynamics acts on the resulting system.

This leads to a sharper question: how should projection behavior be analyzed without conflating manifold geometry, parametrization, and solver-level behavior?

In this paper, we argue that the key difficulty is a previously unrecognized ambiguity at the parametrization layer. Even when the geometric definition of the manifold is fixed, the same local projection problem does not uniquely determine its parametrized form. This obscures solver-dependent behavior and makes it difficult to separate parametrization-level limitations from properties of the underlying manifold.

To address this issue, we develop a projection framework that makes this separation explicit and introduce the BTN parametrization as a parametrization-level reformulation

that resolves the ambiguity. This framework allows projection behavior to be analyzed through its numerical solution by separating the geometric content of the problem from the parametrized system on which the solver acts.

II. LAYERED STRUCTURE OF THE PROJECTION PROBLEM

The Introduction identified a basic interpretive difficulty: projection behavior is often discussed in ways that mix geometric properties of the flowed manifold with features that depend on the chosen parametrization and numerical solver. To separate these roles, we begin by organizing the problem into three layers: geometry, parametrization, and solver. As a concrete reference point within this framework, we use the conventional local (u, λ) projection formulation in flowed-manifold HMC, referred to here as the standard (u, λ) formulation.[14]

In the present discussion, this three-layer organization is used as the analytical framework. The geometry layer fixes the local projection target in z -space through the local normal-footpoint picture shown schematically in Fig. 1(a). The parametrization layer specifies how the same local projection problem is written as a parametrized system. The solver layer describes the numerical iteration acting on the resulting equation system. This layered structure provides the reference used later to change parametrization without changing the underlying geometric projection problem.

A. Geometry layer: projection target on M_τ

Let $x \in \mathbb{R}^N$ evolve under the antiholomorphic flow

$$\frac{dz_i(\tau)}{d\tau} = [\partial_i S(z(\tau))]^*, \quad z_i(0) = x_i, \quad (1)$$

and define the flowed manifold

$$M_\tau = \{z_\tau(x) \mid x \in \mathbb{R}^N\} \subset \mathbb{C}^N. \quad (2)$$

We consider a local projection problem near a base point

$$z_0 = z_\tau(x) \in M_\tau. \quad (3)$$

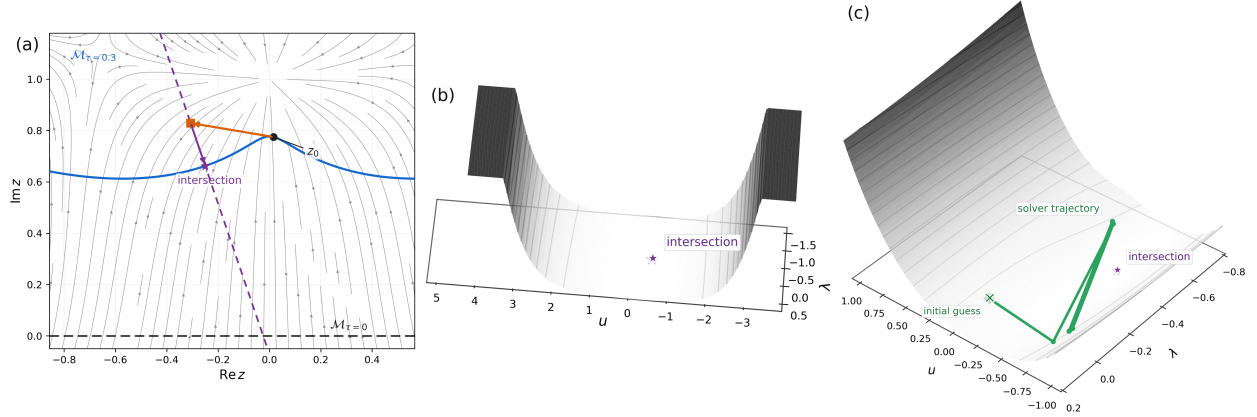


FIG. 1. Three-layer schematic for the standard (u, λ) formulation. (a) Geometry layer: projection is defined in z -space as a local normal-footpoint problem on M_τ . (b) Parametrization layer: for fixed local data, the same local projection problem is rewritten in auxiliary variables, equivalently as the minimization of the induced landscape $\|f_{\text{std}}\|^2$. (c) Solver layer: numerical iteration acting on the parametrized system.

For a nearby trial configuration, write

$$\tilde{z} = z_0 + \delta z. \quad (4)$$

At the geometry layer, the projection target is specified purely in z -space: one seeks a point

$$z^* \in M_\tau \quad (5)$$

such that the correction from the trial configuration to the manifold is normal to M_τ at the base point,

$$\tilde{z} - z^* \in N_{z_0} M_\tau. \quad (6)$$

This local geometric reference is the content of Fig. 1(a). It fixes the projection target that is rewritten in standard (u, λ) variables in the next subsection and kept unchanged when the parametrization layer is replaced in Sec. III.

B. Parametrization layer: the standard (u, λ) formulation

The standard formulation rewrites the local projection problem in auxiliary real variables

$$u, \lambda \in \mathbb{R}^N, \quad w = (u, \lambda) \in \mathbb{R}^{2N}. \quad (7)$$

At a base point $z_\tau(x)$, we introduce the local tangent and normal directions

$$E_\alpha(x) = \frac{\partial z_\tau(x)}{\partial x^\alpha}, \quad F_\alpha(x) = i E_\alpha(x), \quad (8)$$

and write the projection condition as

$$f_{\text{std}}^I(u, \lambda; \tilde{z}, x) = z_\tau^I(x + u) - \tilde{z}^I + F_\alpha^I(x) \lambda^\alpha = 0. \quad (9)$$

In the standard formulation, u and λ are auxiliary real variables introduced to rewrite the local projection problem for fixed local data $(\tilde{z}, x)$. The equation $f_{\text{std}}(u, \lambda; \tilde{z}, x) = 0$ is then the root condition in these auxiliary variables for fixed local data $(\tilde{z}, x)$. Equivalently, the same problem can be viewed as minimizing

$$\|f_{\text{std}}(u, \lambda; \tilde{z}, x)\|^2,$$

so that projected solutions appear as minima of the induced landscape in (u, λ) space.

This parametrization-level structure is summarized in Fig. 1(b). At this layer, the standard (u, λ) formulation should therefore be understood simply as a solver-facing reformulation of the local projection problem in auxiliary variables.

C. Solver layer: numerical behavior on the parametrized problem

Once the parametrized system has been fixed, projection becomes a numerical problem for the chosen solver. In the standard formulation, one solves

$$f_{\text{std}}(w; \tilde{z}, x) = 0, \quad w = (u, \lambda), \quad (10)$$

for the auxiliary variables w . A Newton-type step takes the form

$$w^{(k+1)} = w^{(k)} + \Delta w^{(k)}, \quad J(w^{(k)}) \Delta w^{(k)} = -f_{\text{std}}(w^{(k)}). \quad (11)$$

At this stage, the relevant objects are numerical: the initial guess, iteration path, convergence, and non-convergence. Figure 1(c) illustrates this solver-level structure. Whether iteration reaches a solution, stalls, or fails within tolerance is therefore a statement about the solver acting on the standard parametrized equation.

This completes the standard picture. The geometry layer fixes the local projection target, the parametrization layer determines the parametrized system for the same local projection problem, and the solver layer governs the observed numerical behavior. Section III keeps the same geometry layer and replaces only the parametrization layer with BTN.

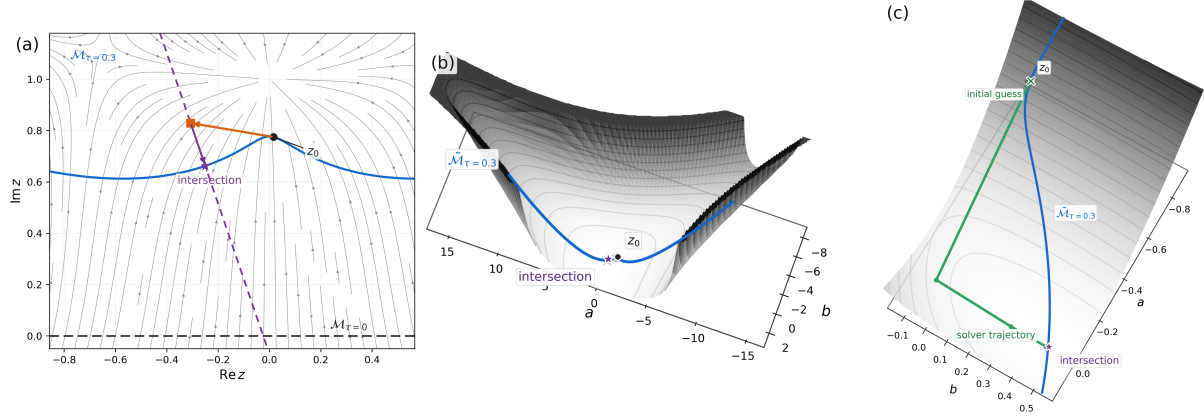


FIG. 2. Three-layer schematic for the BTN parametrization. (a) Geometry layer: the geometric projection target is unchanged. (b) Parametrization layer: BTN makes the manifold condition explicit in the parametrization, so the same local projection problem is reformulated in a parametrized system that exposes the local projection structure explicitly to the solver. (c) Solver layer: numerical trajectory after replacing the parametrization layer with BTN.

III. BTN REFORMULATION OF THE SAME PROJECTION PROBLEM

Section II separated the projection problem into geometry, parametrization, and solver behavior. Within that structure, the geometric target is fixed in z -space, and solver behavior becomes concrete only after a parametrization is chosen. This section reconsiders the parametrization layer.

We begin with a simple numerical observation that motivates this step. We consider the one-dimensional toy model with complex action

$$S(z; a, b) = -i \left(\frac{z^3}{3} + az \right) - \log(z - ib), \quad (12)$$

with $a = b = 1$. As a consistency diagnostic, we use the Schwinger–Dyson observable

$$\mathcal{O}_{\text{SD}}(z) = -i(z - ib)(z^2 + a) - 2, \quad (13)$$

for which

$$\langle \mathcal{O}_{\text{SD}}(z) \rangle = 0 \quad (14)$$

is an exact analytical result in this model. Figure 3 shows seed-level estimates of $\langle \mathcal{O}_{\text{SD}} \rangle$ obtained from a naive implementation based on the standard parametrization with the simplified Newton method. This is not a full numerical comparison. It only establishes a

clear consistency problem: seed means remain systematically displaced from the analytical value.

By itself, this observation does not identify which layer is responsible. The deviation may reflect geometric difficulty of the projection target, the parametrization used to pose the same local projection problem, or the behavior of the numerical solver acting on the resulting parametrized system. It should therefore not be read directly as evidence for any one of these explanations. Before turning to tuning within the standard parametrization or to alternative numerical strategies, we make a different conceptual move. By separating the projection problem into geometry, parametrization, and solver layers, we identify the parametrization layer itself as an independent source of numerical behavior. This allows us to ask, in explicit form, whether the same local projection problem can be reformulated at layer 2 in a way that changes solver behavior without changing the underlying geometry.

This section reformulates the same local projection problem in BTN variables. The point is not to change the manifold or introduce a different projection target, but to reformulate the same projection problem in variables that make the manifold condition explicit. Figure 2 summarizes the corresponding three-layer picture.

A. Why the parametrization layer needs to be changed

Figure 2(a) fixes the geometric reference for this reformulation. The issue is not the projection target in z -space, but its formulation at the parametrization layer.

In the standard (u, λ) formulation, the projection problem is posed for fixed local data as a solver-facing minimization problem in auxiliary variables. The question is whether the same local projection problem can be reformulated in variables that make the manifold condition explicit.

BTN is introduced as a parametrization-level reformulation of the same local projection problem. The essential change is that BTN makes the manifold condition itself explicit in the parametrization, so the local projection structure is presented to the solver in a different form.

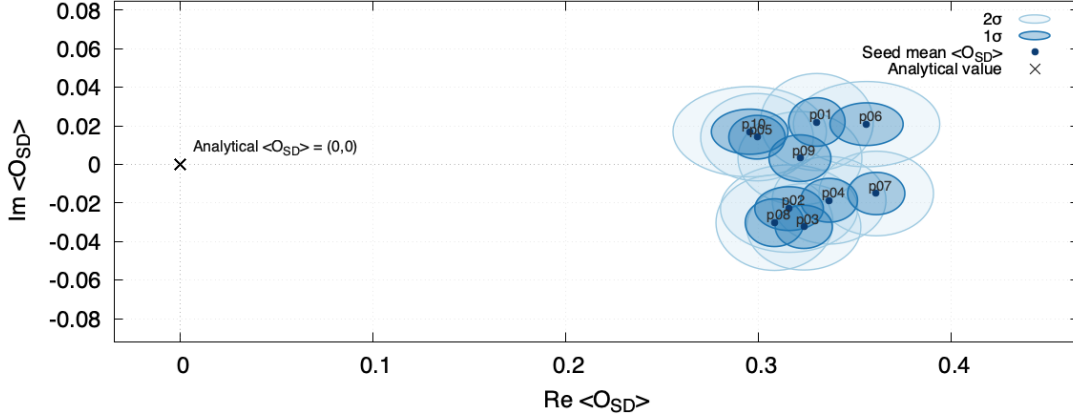


FIG. 3. Motivating consistency deviation in the standard parametrization with the simplified Newton method. The figure shows seed-level estimates of $\langle \mathcal{O}_{SD} \rangle$ in the complex plane, together with their 1σ and 2σ regions. The analytical value is $(0,0)$. The role of this figure is only to show that a substantial consistency deviation can already arise under a naive implementation in the standard parametrization. It is not intended as a full numerical comparison, which is deferred to Sec. IV.

B. BTN parametrization and the explicit manifold condition

We now introduce the backflow tangent-normal (BTN) parametrization as an alternative parametrization-level reformulation of the same local projection problem. Its defining object is the backflow map

$$f_{\text{BTN}}(z) = \Phi_{-\tau}(z), \quad (15)$$

where Φ_s denotes the flow map generated by the antiholomorphic flow,

$$\frac{d}{ds} \Phi_s^i(z) = [\partial_i S(\Phi_s(z))]^*, \quad \Phi_0^i(z) = z^i. \quad (16)$$

Thus f_{BTN} maps a configuration backward along the flow by flow time τ .

If $z \in M_\tau$, then $z = \Phi_\tau(x)$ for some $x \in \mathbb{R}^N$, and therefore

$$f_{\text{BTN}}(z) = \Phi_{-\tau}(z) = x \in \mathbb{R}^N. \quad (17)$$

Accordingly, in BTN variables

$$\xi = f_{\text{BTN}}(z), \quad (18)$$

the flowed manifold is characterized locally by the reality condition

$$\Im \xi = 0. \quad (19)$$

This condition makes the manifold constraint explicit in the BTN variables and is the basis for the projection formulation in the next subsection. Figure 2(b) illustrates this parametrization-level reformulation. The next subsection writes the corresponding projection problem explicitly in terms of the (a, b) variables associated with the local tangent and normal directions.

C. Projection problem in BTN form

The BTN parametrization becomes explicit once the local tangent–normal basis at $z_\tau(x)$ is introduced:

$$E_\alpha(x) = \frac{\partial z_\tau(x)}{\partial x^\alpha}, \quad F_\alpha(x) = i E_\alpha(x). \quad (20)$$

For the same base point $z_\tau(x)$, we write the trial configuration locally as

$$\tilde{z} = z_\tau(x) + \delta z. \quad (21)$$

The variables a and b describe the tangent and normal components of the local displacement from $\tilde{z}$, respectively. The projection problem is then written in BTN form as the coupled system

$$\mathcal{C}(a, b; \tilde{z}, x) = \Im f_{\text{BTN}}(\tilde{z} - E_\alpha(x)a^\alpha - F_\alpha(x)b^\alpha) = 0, \quad a = 0. \quad (22)$$

Thus, in BTN variables, the same local projection problem for fixed local data $(\tilde{z}, x)$ is posed by the above coupled system.

Fig. 2(b) illustrates this parametrization-level formulation. The next subsection considers the numerical solver acting on this BTN system.

D. Solver behavior after changing the parametrization

At the solver layer, the numerical task remains to solve a nonlinear projection system. What changes is not the role of the solver, but the parametrized equation on which it acts.

For a base point $z_\tau(x)$, we write the trial configuration as

$$\tilde{z} = z_\tau(x) + \delta z. \quad (23)$$

In the standard formulation, the solver acts on

$$f_{\text{std}}(u, \lambda; \tilde{z}, x) = 0, \quad (24)$$

whereas after the layer-2 reformulation it acts on the BTN system

$$\mathcal{C}(a, b; \tilde{z}, x) = 0, \quad a = 0. \quad (25)$$

The resulting numerical behavior is therefore a response to two different parametrized systems for the same local projection problem.

This is where parametrization-level differences acquire numerical significance. Any improved convergence or reduced instability, as illustrated in Fig. 2(c), should therefore be interpreted as solver behavior on a different equation system, not as a change in the geometric projection target. This provides the interpretive basis for the numerical results presented in the next section.

IV. NUMERICAL RESULTS

We now turn to the numerical comparison implied by the framework developed in Secs. II and III. The same local projection problem for fixed local data $(\tilde{z}, x)$ is used throughout, but its practical treatment depends on the chosen parametrization and solver behavior. This section compares computational setups for that same local projection problem and interprets the resulting differences.

A. Numerical Setup and Geometry Context

The comparisons are carried out between two implemented computational setups for the same local projection problem. In the fallback-disabled setup, projection is solved only in

the standard parametrization using the simplified Newton method. This choice is taken as the standard-side baseline because simplified Newton is already used in the flowed-manifold HMC literature as a practical solver for projection constraints.[14] In the fallback-enabled setup, the same standard solve is attempted first, and the computation switches to the BTN system only after failure, with the implemented rescue policy, gating rules, and iteration-cap criteria specified in Appendix B. The complete seed-level raw data used in the numerical comparison are listed in Appendix A. At that fallback stage, we use DFOLS as a derivative-free nonlinear least-squares solver based on a trust-region strategy, implemented without explicit Jacobian evaluation.[15, 16] The comparison is therefore between two implemented computational setups, not a full cross-combination study of parametrizations and solvers.

We consider the one-dimensional toy model with complex action

$$S(z; a, b) = -i \left(\frac{z^3}{3} + az \right) - \log(z - ib),$$

with $a = b = 1$. The simulation controls are the flow time τ , the HMC trajectory length L , and the HMC integration step count n_{step} . Unless stated otherwise, each run uses 24 chains with 50,000 configurations per chain. The principal paired comparison is carried out at

$$(\tau, L, n_{\text{step}}) = (0.30, 2, 20).$$

The principal paired comparison uses common random seeds, matched initial conditions, and matched simulation controls.

As a consistency diagnostic, we use the Schwinger–Dyson observable

$$\mathcal{O}_{\text{SD}}(z) = -i(z - ib)(z^2 + a) - 2. \quad (26)$$

For this model, the Schwinger–Dyson relation

$$\langle \mathcal{O}_{\text{SD}}(z) \rangle = 0 \quad (27)$$

is an exact analytical result. The pass fractions P_{68} and P_{95} denote the fractions of runs whose estimated value of $\langle \mathcal{O}_{\text{SD}}(z) \rangle$ is consistent with zero at the 1σ and 2σ levels, respectively. At fixed parameter setting, F denotes the total number of projection failures summed over all chains.

Figure 4 provides the geometric context for these experiments in the $(\text{Re } z, \text{Im } z)$ plane. It shows the undeformed contour, the relevant saddle points, and the representative flowed

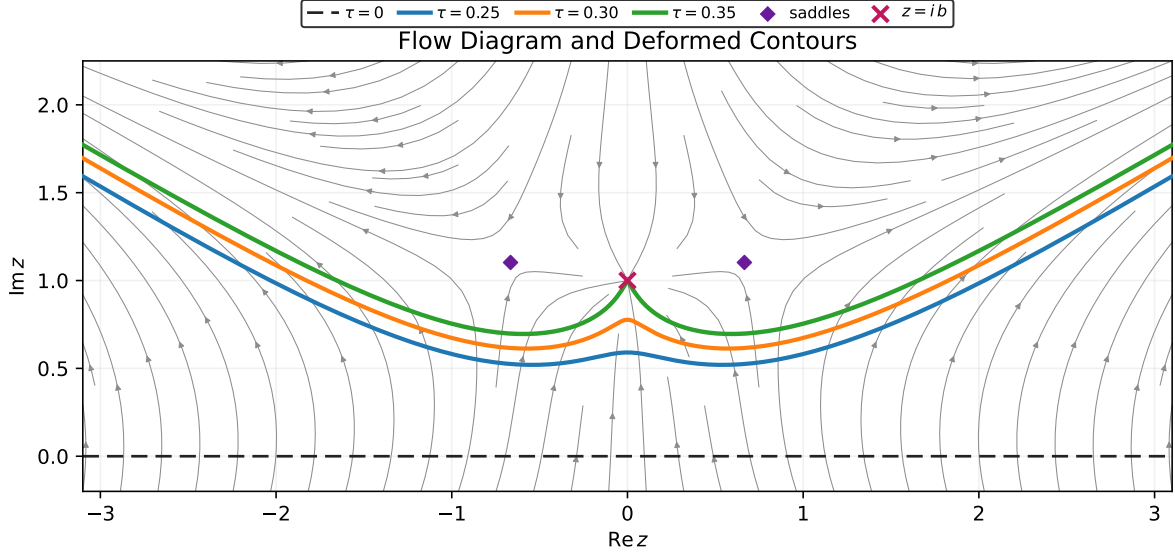


FIG. 4. Toy-model geometry at $a = b = 1$ under the antiholomorphic flow. The figure shows the flow diagram generated by $dz/d\tau = \overline{\partial_z S}$, together with the undeformed contour (the real axis), the relevant saddle points with $\text{Im } z > 0$, and the three flowed contours used in the numerical experiments ($\tau = 0.25, 0.30, 0.35$). For $\tau = 0.35$, the contour is represented by two smooth branches terminating at $z = ib$, reflecting the discontinuity induced by the singularity.

contours used in the numerical study. The figure serves only to locate the numerical tests within the toy-model geometry; the comparisons reported below are made between computational setups for the same local projection problem.

B. Principal Comparison at the Principal Setting

Table I gives the principal paired comparison under matched controls. The key feature is the combined pattern across diagnostics. The fallback-enabled setup attains substantially higher Schwinger–Dyson consistency pass fractions while reducing the projection-failure burden to zero, and this improvement is not accompanied by an appreciable runtime increase because mean runtimes remain nearly identical. At the principal setting, the distinction is therefore the simultaneous improvement of consistency and failure burden at comparable runtime. This makes the principal comparison the clearest numerical separation between

TABLE I. Principal paired comparison of computational setups under fixed controls.

Setup	N_{seeds}	P_{68}	P_{95}	F	Mean runtime (s)
Fallback-enabled	10	0.90	1.00	0	443.766
Fallback-disabled	10	0.35	0.50	252841	444.964

the two setups in the present study.

1. Step-count Dependence within Each Setup

At fixed $(\tau, L) = (0.30, 2)$, Fig. 5 shows the dependence on n_{step} within each computational setup.

For the fallback-disabled setup, increasing n_{step} improves both diagnostics. The 2σ pass fraction rises from 0.50 at $n_{\text{step}} = 10$ to 1.00 at $n_{\text{step}} = 40$, while the projection failure burden decreases from 809660 to 7119. Thus, a larger step count clearly mitigates the numerical difficulty of the fallback-disabled setup, although its failure burden remains nonzero throughout the scan.

For the fallback-enabled setup, by contrast, the projection-failure burden remains zero across the full scan, while P_{95} stays in the high-consistency range. The small variation in P_{95} is consistent with finite-sample fluctuation and does not alter the overall pattern.

This comparison is useful because it separates step-count tuning within a fixed setup from setup-level differences. Increasing n_{step} improves the fallback-disabled setup, but does not remove the contrast with the fallback-enabled setup in projection-failure behavior.

2. Tuning versus Fallback Policy

Table II compares two non-identical setups: the fallback-enabled setup at $n_{\text{step}} = 20$ and the fallback-disabled setup at $n_{\text{step}} = 40$. Because the step counts are different, this is not a strict paired comparison under matched controls. Its role is instead to provide an efficiency cross-check of whether step-count tuning alone can compensate for the absence of fallback.

The comparison shows that step-count tuning can substantially improve the fallback-disabled setup. At $n_{\text{step}} = 40$, it reaches the same $P_{95} = 0.975$ as the fallback-enabled setup at $n_{\text{step}} = 20$, while the corresponding P_{68} values also lie at a comparable high-

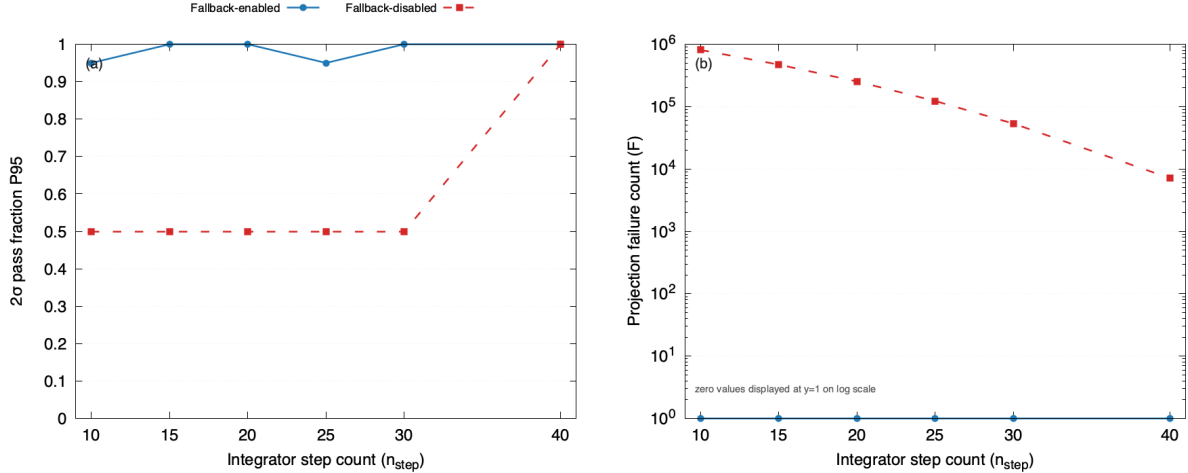


FIG. 5. Step-count dependence within each computational setup on the paired 10-seed basis. Panel (a): P_{95} versus n_{step} . Panel (b): projection failure count F versus n_{step} on a logarithmic axis. For the fallback-enabled setup, zero values are displayed at $y = 1$ on the log scale.

TABLE II. Efficiency cross-check between step-count tuning and fallback policy.

Setup	P_{68}	P_{95}	F	Mean runtime (s)
Fallback-enabled, $n_{\text{step}} = 20$	0.85	0.975	0	440.112
Fallback-disabled, $n_{\text{step}} = 40$	0.80	0.975	14171	639.5005

consistency level. What remains structurally distinct is projection-failure behavior and computational cost: the fallback-disabled setup still has a nonzero failure burden $F = 14171$ and a substantially larger mean runtime.

This cross-check therefore complements the principal paired comparison. Increasing n_{step} can recover part of the consistency performance of the fallback-disabled setup, but does not remove the separation in failure burden and runtime shown in Table II.

C. Flow-Time Dependence

At fixed $(L, n_{\text{step}}) = (2, 20)$, Fig. 6 summarizes how the numerical behavior changes with flow time.

The figure shows a clear progression from a low-difficulty regime to a hard regime as τ increases. At $\tau = 0.25$, both setups exhibit high P_{95} and zero projection failure burden. A clear separation then appears at $\tau = 0.30$: the fallback-enabled setup remains in the

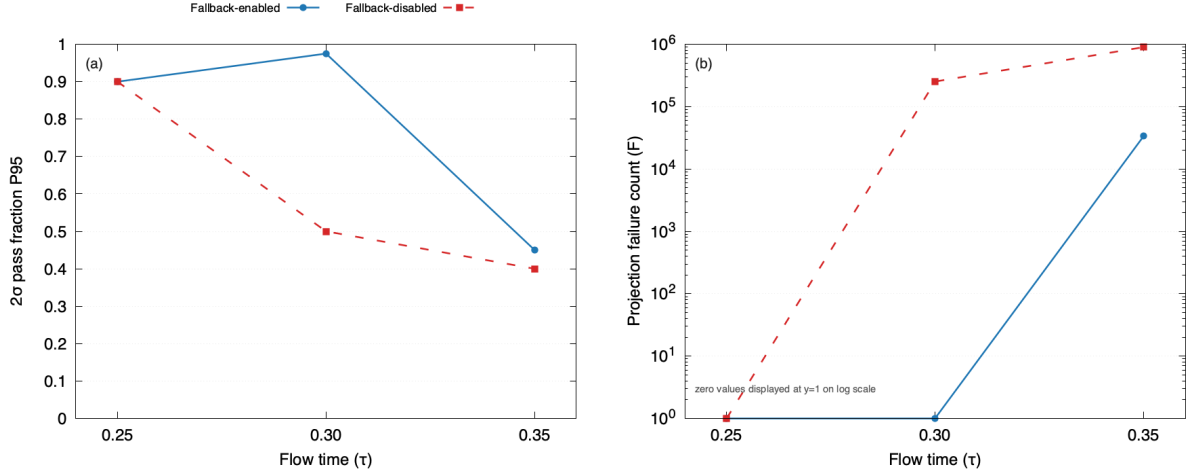


FIG. 6. Flow-time dependence under fixed controls. Panel (a): P_{95} versus flow time τ . Panel (b): projection failure count F versus flow time τ on a logarithmic axis. Zero values are displayed at $y = 1$ on the log scale.

high-consistency, zero-failure regime, whereas the fallback-disabled setup already shows a substantial drop in P_{95} together with a large nonzero failure burden. By $\tau = 0.35$, both setups deteriorate, indicating that the projection problem has entered a substantially harder numerical regime.

This scan also clarifies how the comparison changes across regimes. The fallback is most effective in delaying the onset of severe projection difficulty as τ increases, and even at $\tau = 0.35$ the two setups remain clearly separated in projection-failure burden. The point is therefore not simply that both setups become difficult, but that the comparison enters a regime in which the remaining failures themselves become the relevant object of analysis.

This motivates the next subsection. Once both setups have entered the hard regime, the question is no longer only whether projection failures occur, but how the remaining failures are distributed across seeds and how they relate to the seed-level bias diagnostic.

D. Hard-Regime Diagnostics

At $(\tau, L, n_{\text{step}}) = (0.35, 2, 20)$, both setups have entered the hard regime identified in the previous subsection. The relevant question is therefore not only whether projection failures occur, but how the remaining failures are organized across seeds. To probe this structure, we

first introduce two seed-level diagnostics: the failure count F_p and the normalized Schwinger–Dyson bias diagnostic B_p . A setup-aggregated localization diagnostic R_{loc} is then introduced below as supplementary evidence.

For each seed p , let F_p denote the projection-failure count for that seed. We use

$$B_p = \frac{\langle \mathcal{O}_{\text{SD}} \rangle_p}{\text{err}[\langle \mathcal{O}_{\text{SD}} \rangle_p]}, \quad (28)$$

as a seed-level normalized Schwinger–Dyson bias diagnostic, where $\text{err}[\langle \mathcal{O}_{\text{SD}} \rangle_p]$ is obtained by jackknife.

Figure 7 shows that the hard-regime difference is not exhausted by the amount of remaining failure alone. The two setups are already clearly separated on the F_p axis, with the fallback-enabled seeds occupying a substantially lower failure-count range than the fallback-disabled seeds. At the same time, the marginal ranges of B_p still overlap. The main distinction therefore lies not in the bias range alone, but in how failure count and bias diagnostic are organized jointly across seeds.

This distinction is reflected in the Spearman correlation between F_p and B_p , which measures the monotonic association between the seed-level failure count and the normalized bias diagnostic. For the fallback-disabled setup, the Spearman correlation is positive,

$$\text{corr}(F_p, B_p) = 0.7576,$$

whereas for the fallback-enabled setup it is negative,

$$\text{corr}(F_p, B_p) = -0.6242.$$

Thus, even within the hard regime, changing setup affects not only the aggregate amount of remaining failure but also the seed-level bias–failure structure.

The setup-aggregated quantity R_{loc} provides a supplementary check on this picture. We define

$$R_{\text{loc}} = \frac{1}{N_{\text{fail}}} \sum_{z \in \mathcal{F}} |\text{Re } z|, \quad (29)$$

where $\mathcal{F}$ is the set of all projection-failure points from all seeds in a given setup, and $N_{\text{fail}} = |\mathcal{F}|$. The fallback-enabled setup gives

$$R_{\text{loc}} = 0.003262,$$

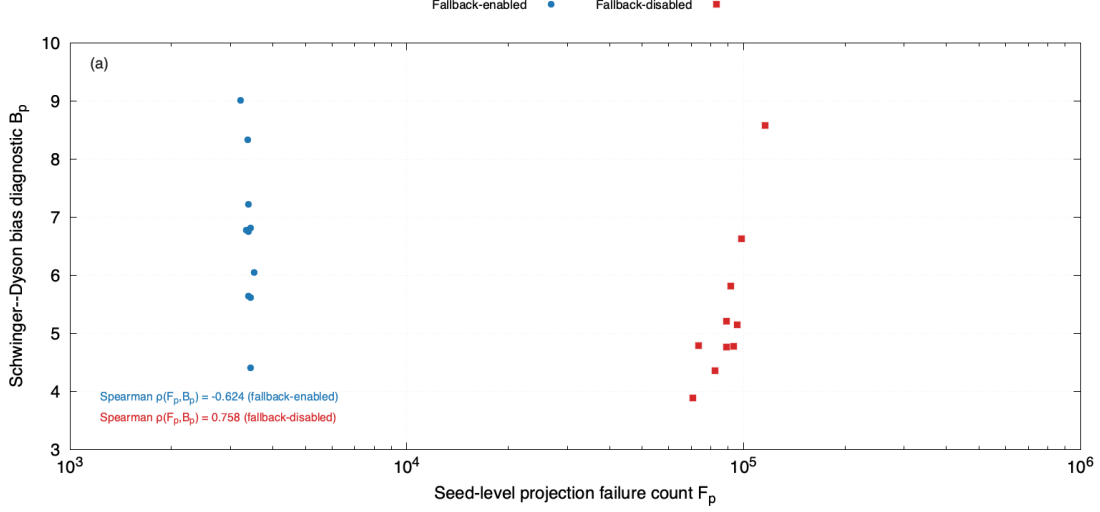


FIG. 7. Hard-regime seed-level bias–failure structure at $\tau = 0.35$. Scatter plot of the normalized Schwinger–Dyson bias diagnostic B_p versus the seed-level projection failure count F_p for the two computational setups. The Spearman correlation within each computational setup is indicated in the panel, and the horizontal axis is shown on a logarithmic scale.

TABLE III. Hard-regime setup-aggregated diagnostic summary at $\tau = 0.35$.

Setup	N_{seeds}	$\text{corr}(F_p, B_p)$	R_{loc}
Fallback-enabled	10	-0.6242	0.003262
Fallback-disabled	10	0.7576	0.061150

whereas the fallback-disabled setup gives

$$R_{loc} = 0.061150.$$

Since the singularity is located at $z = ib$, smaller R_{loc} means smaller typical horizontal displacement from the singular line $\text{Re } z = 0$, and is therefore consistent with a more localized failure pattern near the singularity. Because this is an indirect diagnostic based on prior knowledge of singularity location, we use it as supporting evidence rather than as the primary basis of the argument.

Taken together, Fig. 7 and Table III show that the remaining hard-regime failures still carry setup-dependent structure. The difference persists not only in aggregate projection-failure burden, but also in the seed-level relation between failure count and bias diagnostic.

V. DISCUSSION

The numerical comparison in Sec. IV supports a specific interpretation of projection behavior. Within the framework of Secs. II and III, the relevant object is the geometry-to-solver interface. More concretely, this interface is the parametrized system on which the solver acts, carrying the geometric information relevant to the same local projection problem. BTN changes this interface while keeping the geometric problem fixed, so the observed differences in correctness behavior and projection-failure burden should be read as consequences of an interface change under a fixed projection target.

This effect is not reducible to tuning within a fixed setup. Increasing n_{step} improves Schwinger–Dyson consistency and reduces the projection-failure burden in the fallback-disabled setup, but does not reproduce the qualitative behavior obtained after the interface change. In particular, matching the high-consistency level still leaves a nonzero failure burden and requires substantially larger runtime. The distinction is therefore between two parametrizations of the same local projection problem presented to the solver, not between tuned and untuned versions of a single setup.

The same distinction persists in the hard regime. Although both setups deteriorate at large flow time, the difference between them does not disappear. The projection-failure burden remains different, and more importantly, the seed-level relation between the failure count F_p and the bias diagnostic B_p changes sign across setups. This shows that the remaining failures are organized differently, not merely reduced in number.

The localization diagnostic R_{loc} provides additional, indirect support for this picture, with smaller values in the fallback-enabled case indicating a more localized failure pattern near the singular line. Taken together, these observations show that even in the hard regime, projection behavior retains setup-dependent structure.

The broader implication is interpretive. Projection failure should be interpreted through the parametrized interface by which the solver accesses the problem, rather than as a direct readout of manifold geometry. Geometry fixes the target, but observed numerical behavior is shaped by the parametrization layer and the solver acting on it. Projection behavior therefore becomes interpretable only after separating the geometric target from its numerical interface.

VI. CONCLUSION

The main conclusion of this work is that projection behavior in practice is controlled by the geometry-to-solver interface. Within the present framework, BTN changes that interface by changing the parametrized system on which the solver acts, while keeping the geometric target fixed. Under the tested computational setups, this interface change produces a clear numerical separation in correctness behavior and projection-failure burden.

This conclusion is supported by two features of the numerical comparison. First, the observed improvement is not reducible to step-count tuning within the fallback-disabled setup: although increasing n_{step} mitigates numerical difficulty, it does not reproduce the combined behavior obtained after the interface change. Second, even in the hard regime, where both setups become more difficult, the remaining failures do not become indistinguishable across setups. The aggregate projection-failure burden still differs, and more importantly, the seed-level relation between F_p and B_p changes sign across setups. These observations do not establish a general claim beyond the tested setting, but they show that the numerical consequences of changing the parametrized interface are measurable and remain structurally visible under the tested computational setups.

This interface-based viewpoint also suggests a natural direction for further work. Although the present study is carried out in a controlled low-dimensional setting, the model already captures an important structural feature through singular behavior associated with the determinant-like factor in the action. A natural next step is to test the same geometry–parametrization–solver separation in more physical models and higher-dimensional systems. Related thimble-based methods, such as the tempered Lefschetz thimble method [17] and its worldvolume extension [18], are also formulated in terms of constrained numerical evolution on flowed manifolds or their extensions. They therefore provide natural settings in which to examine how broadly the geometry-to-solver interface controls projection behavior.

Appendix A: Complete seed-level raw data

This appendix lists the complete seed-level raw data underlying the numerical comparison in Sec. IV. Each row corresponds to a single simulation instance at fixed $(\tau, L, n_{\text{step}}, N_{\text{chain}}, s)$ and a specified computational setup. No processed summaries or derived quantities are

introduced here.

The columns are as follows: τ (flow time), L (system size), n_{step} (number of MD steps per trajectory), N_{chain} (number of configurations per chain), s (seed index), setup (fallback-enabled or fallback-disabled), $\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$, $\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$, their corresponding statistical errors σ_{Re} and σ_{Im} , the projection-failure count F , and the runtime T .

1. Data at $L = 2$, $N_{\text{chain}} = 50000$

a. $\tau = 0.25$

TABLE IV: Complete seed-level raw data at $\tau = 0.25$, $L = 2$,
 $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.25	20	p01	fallback-disabled	-0.01597761	0.02426025	0.04287129	0.03056086	0	300.74
0.25	20	p01	fallback-enabled	-0.01597761	0.02426025	0.04287129	0.03056086	0	300.4
0.25	20	p02	fallback-disabled	0.005363867	0.02287634	-0.08090011	0.03115546	0	301.74
0.25	20	p02	fallback-enabled	0.005363867	0.02287634	-0.08090011	0.03115546	0	301.53
0.25	20	p03	fallback-disabled	0.008103256	0.0235761	0.03171359	0.03352095	0	302.54
0.25	20	p03	fallback-enabled	0.008103256	0.0235761	0.03171359	0.03352095	0	303.86
0.25	20	p04	fallback-disabled	-0.0180006	0.02683924	0.03331719	0.03181073	0	298.96
0.25	20	p04	fallback-enabled	-0.0180006	0.02683924	0.03331719	0.03181073	0	299.77
0.25	20	p05	fallback-disabled	-0.01703621	0.02340603	0.04467479	0.02917362	0	303.67
0.25	20	p05	fallback-enabled	-0.01703621	0.02340603	0.04467479	0.02917362	0	304.02
0.25	20	p06	fallback-disabled	0.02469963	0.02245425	-0.009905995	0.03054014	0	301.86
0.25	20	p06	fallback-enabled	0.02469963	0.02245425	-0.009905995	0.03054014	0	303.45
0.25	20	p07	fallback-disabled	-0.01894619	0.02305937	0.07410036	0.02960689	0	299.31
0.25	20	p07	fallback-enabled	-0.01894619	0.02305937	0.07410036	0.02960689	0	302.92
0.25	20	p08	fallback-disabled	0.02314462	0.02194945	0.0321442	0.0349237	0	303.92
0.25	20	p08	fallback-enabled	0.02314462	0.02194945	0.0321442	0.0349237	0	300.57
0.25	20	p09	fallback-disabled	-0.005809142	0.02292086	-0.02657586	0.03109853	0	300.88
0.25	20	p09	fallback-enabled	-0.005809142	0.02292086	-0.02657586	0.03109853	0	302.96

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.25	20	p10	fallback-disabled	-0.03094306	0.02807852	-0.0268913	0.03014611	0	302.96
0.25	20	p10	fallback-enabled	-0.03094306	0.02807852	-0.0268913	0.03014611	0	299.75

b. $\tau = 0.30$

$n_{\text{step}} = 10$

TABLE V: Complete seed-level raw data at $\tau = 0.30$, $L = 2$,
 $n_{\text{step}} = 10$, $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	10	p01	fallback-disabled	0.4244965	0.02948404	-0.006577985	0.02696228	81586	486.48
0.30	10	p01	fallback-enabled	-0.02005242	0.02223289	0.02291615	0.01791569	0	482
0.30	10	p02	fallback-disabled	0.4430117	0.02862381	-0.0143745	0.02389634	81398	496.57
0.30	10	p02	fallback-enabled	-0.01141924	0.0245714	-0.0006116222	0.01519675	0	480.74
0.30	10	p03	fallback-disabled	0.409462	0.03323803	0.03297957	0.02198213	80752	487.08
0.30	10	p03	fallback-enabled	-0.003030307	0.02349906	0.02283625	0.01803818	0	488.88
0.30	10	p04	fallback-disabled	0.3610631	0.04106622	-0.001727788	0.02236933	80485	485.17
0.30	10	p04	fallback-enabled	0.003978661	0.02658498	0.02417438	0.01569736	0	473.68
0.30	10	p05	fallback-disabled	0.4267979	0.03017775	-0.00511572	0.02360467	81086	484.4
0.30	10	p05	fallback-enabled	-0.04763938	0.02260698	0.0100047	0.01527837	0	483.33
0.30	10	p06	fallback-disabled	0.3903498	0.02875387	0.02151358	0.02248123	80714	478.6
0.30	10	p06	fallback-enabled	0.02300923	0.02291385	0.006921337	0.0166952	0	476.13
0.30	10	p07	fallback-disabled	0.3690355	0.02806046	-0.01479928	0.02116501	80522	486.09
0.30	10	p07	fallback-enabled	0.021111223	0.02215033	-0.00005290005	0.01782402	0	478.86
0.30	10	p08	fallback-disabled	0.395095	0.03546897	-0.01554222	0.02225928	80632	491.45
0.30	10	p08	fallback-enabled	-0.008840558	0.02260125	0.001913235	0.01766545	0	492.5
0.30	10	p09	fallback-disabled	0.3701778	0.03114223	-0.0167189	0.0244602	80997	487.49
0.30	10	p09	fallback-enabled	-0.003665546	0.02249193	0.004222425	0.02097319	0	482.13
0.30	10	p10	fallback-disabled	0.4761485	0.03113438	0.02015561	0.02382918	81488	489.26
0.30	10	p10	fallback-enabled	0.007968695	0.02418534	0.002543118	0.01661499	0	486.35

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
--------	-------------------	-----	-------	---	----------------------	---	----------------------	-----	-----

$n_{\text{step}} = 15$

TABLE VI: Complete seed-level raw data at $\tau = 0.30$, $L = 2$,
 $n_{\text{step}} = 15$, $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	15	p01	fallback-disabled	0.4045691	0.02868714	-0.0177	0.02293918	46750	388.02
0.30	15	p01	fallback-enabled	-0.007569597	0.02102112	-0.01277903	0.01873268	0	395.65
0.30	15	p02	fallback-disabled	0.3944457	0.03715209	-0.01152983	0.02257399	46935	394.64
0.30	15	p02	fallback-enabled	-0.0006829727	0.02273645	0.01806368	0.01630947	0	393.93
0.30	15	p03	fallback-disabled	0.3888643	0.03330069	-0.01453448	0.02271203	47096	390.39
0.30	15	p03	fallback-enabled	-0.02305737	0.02837118	0.001631587	0.01759407	0	391.73
0.30	15	p04	fallback-disabled	0.3977785	0.02943864	-0.02212757	0.02354846	47445	392.6
0.30	15	p04	fallback-enabled	0.02380581	0.02451722	0.006161726	0.01706006	0	395.97
0.30	15	p05	fallback-disabled	0.4578971	0.03026821	0.02611736	0.02799136	47322	388.09
0.30	15	p05	fallback-enabled	0.002461281	0.02189155	0.01994369	0.02245162	0	395.57
0.30	15	p06	fallback-disabled	0.4184747	0.02845521	-0.01877132	0.02352734	46820	389.31
0.30	15	p06	fallback-enabled	0.0190451	0.02154219	-0.009222727	0.01619903	0	398.99
0.30	15	p07	fallback-disabled	0.4210135	0.02958873	0.01157456	0.02242569	47478	395.96
0.30	15	p07	fallback-enabled	0.01219659	0.02055371	-0.006531263	0.01896606	0	398.47
0.30	15	p08	fallback-disabled	0.3809394	0.02868057	-0.02443243	0.02484808	46957	388.19
0.30	15	p08	fallback-enabled	0.01889362	0.02234864	-0.02514936	0.01685659	0	397.1
0.30	15	p09	fallback-disabled	0.425061	0.03044056	0.0171224	0.02627556	47500	393.58
0.30	15	p09	fallback-enabled	-0.002118485	0.02259841	0.01877018	0.01684242	0	395.95
0.30	15	p10	fallback-disabled	0.3751506	0.03063771	-0.001982228	0.02298906	47023	389.64
0.30	15	p10	fallback-enabled	0.02961291	0.02491092	0.02898133	0.0169323	0	392.42

$n_{\text{step}} = 20$

TABLE VII: Complete seed-level raw data at $\tau = 0.30$, $L = 2$, $n_{\text{step}} = 20$, $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	20	p01	fallback-disabled	0.3301367	0.02928631	0.02204588	0.02514245	25216	438.28
0.30	20	p01	fallback-enabled	-0.03443686	0.02490557	0.006067888	0.01909925	0	438.73
0.30	20	p02	fallback-disabled	0.3157442	0.03559817	-0.02286679	0.02270146	25159	447.23
0.30	20	p02	fallback-enabled	0.01316177	0.02340605	-0.008462006	0.01728095	0	444.59
0.30	20	p03	fallback-disabled	0.3233599	0.02962957	-0.03234624	0.02238491	25297	443.79
0.30	20	p03	fallback-enabled	-0.02265792	0.02274335	0.00009213191	0.01719349	0	443.46
0.30	20	p04	fallback-disabled	0.3364988	0.02950406	-0.01855954	0.02271296	25650	445.09
0.30	20	p04	fallback-enabled	0.0002867775	0.02220833	0.0008492352	0.0172468	0	441.85
0.30	20	p05	fallback-disabled	0.2990867	0.02921489	0.0141555	0.02265478	24982	446
0.30	20	p05	fallback-enabled	-0.0355874	0.02091242	-0.01360066	0.0174648	0	445.53
0.30	20	p06	fallback-disabled	0.35594	0.03797114	0.02091434	0.02213308	25447	448.81
0.30	20	p06	fallback-enabled	0.002896543	0.02312407	0.01306736	0.0173552	0	444.94
0.30	20	p07	fallback-disabled	0.3609525	0.02971148	-0.01504066	0.02195047	25521	447.61
0.30	20	p07	fallback-enabled	0.003677245	0.02384267	0.002515878	0.0172191	0	443.2
0.30	20	p08	fallback-disabled	0.3081567	0.03033441	-0.03017815	0.02475969	25101	446.11
0.30	20	p08	fallback-enabled	-0.01251869	0.0210087	0.001104682	0.01671093	0	446.42
0.30	20	p09	fallback-disabled	0.3213781	0.03216152	0.003265215	0.02421932	25323	445.61
0.30	20	p09	fallback-enabled	0.008137392	0.02442	0.00759227	0.01825041	0	443.59
0.30	20	p10	fallback-disabled	0.2953126	0.03980594	0.01700489	0.02340573	25145	441.11
0.30	20	p10	fallback-enabled	-0.006499533	0.02219602	0.01227313	0.01793117	0	445.35
0.30	20	p11	fallback-enabled	0.04572318	0.02549845	0.004435595	0.02039795	0	422.72
0.30	20	p12	fallback-enabled	-0.001226566	0.02071121	-0.007563354	0.02010405	0	442.08
0.30	20	p13	fallback-enabled	-0.05402255	0.02287597	0.01636651	0.01643316	0	400.08
0.30	20	p14	fallback-enabled	-0.01299077	0.01998826	-0.002811184	0.01732319	0	422.73
0.30	20	p15	fallback-enabled	0.01452731	0.02285173	0.008700878	0.02114933	0	442.77
0.30	20	p16	fallback-enabled	-0.02277809	0.02071688	-0.002472182	0.01699219	0	453.69
0.30	20	p17	fallback-enabled	-0.003920603	0.02383609	-0.01335684	0.01769655	0	462.87

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	20	p18	fallback-enabled	-0.02635502	0.02192003	0.0003174896	0.01677419	0	422.7
0.30	20	p19	fallback-enabled	-0.01267806	0.02150406	0.005621156	0.02036192	0	450.8
0.30	20	p20	fallback-enabled	0.0036973	0.02185941	0.0006733488	0.01764751	0	444.14

$n_{\text{step}} = 25$

TABLE VIII: Complete seed-level raw data at $\tau = 0.30$, $L = 2$, $n_{\text{step}} = 25$, $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	25	p01	fallback-disabled	0.2033192	0.02707104	0.01934863	0.02130803	12159	507.64
0.30	25	p01	fallback-enabled	-0.01224741	0.02159386	0.02068645	0.01718252	0	505.62
0.30	25	p02	fallback-disabled	0.2245267	0.02647697	-0.014531	0.02171592	12043	504.2
0.30	25	p02	fallback-enabled	-0.007148679	0.02214834	-0.0218118	0.01635588	0	507.35
0.30	25	p03	fallback-disabled	0.1875988	0.02910184	0.005428313	0.0236658	12070	501.72
0.30	25	p03	fallback-enabled	-0.01046656	0.02441114	-0.02504711	0.02105996	0	505.84
0.30	25	p04	fallback-disabled	0.2036978	0.02644038	-0.01424063	0.0220759	12421	502.82
0.30	25	p04	fallback-enabled	-0.04514615	0.02703394	-0.02577727	0.0168193	0	502.74
0.30	25	p05	fallback-disabled	0.2469445	0.02902859	0.007911469	0.02229814	12376	502.82
0.30	25	p05	fallback-enabled	0.008098718	0.02784851	0.005049693	0.01910769	0	505.8
0.30	25	p06	fallback-disabled	0.2222211	0.02569055	0.00835993	0.02191025	12211	504.93
0.30	25	p06	fallback-enabled	0.0181318	0.02248481	0.03879673	0.01677798	0	504.22
0.30	25	p07	fallback-disabled	0.2132321	0.02755598	-0.001520542	0.02299497	12397	508.22
0.30	25	p07	fallback-enabled	0.01368469	0.02404314	-0.01305452	0.0186704	0	504.67
0.30	25	p08	fallback-disabled	0.1914157	0.02690156	-0.002214502	0.02126138	12375	506.82
0.30	25	p08	fallback-enabled	-0.01182554	0.02269704	0.002325568	0.01753308	0	506.46
0.30	25	p09	fallback-disabled	0.1623738	0.02563022	-0.01081368	0.02204769	12261	504.95
0.30	25	p09	fallback-enabled	-0.02692106	0.02853234	0.01313921	0.01699222	0	503.27
0.30	25	p10	fallback-disabled	0.2308578	0.03506805	-0.008955367	0.0235127	12569	503.1
0.30	25	p10	fallback-enabled	-0.02351595	0.02175878	0.0004279372	0.0178239	0	504.28

τ	n_{step}	s	Setup	$\text{Re } \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im } \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
--------	-------------------	-----	-------	--	----------------------	--	----------------------	-----	-----

$n_{\text{step}} = 40$

TABLE IX: Complete seed-level raw data at $\tau = 0.30$, $L = 2$,
 $n_{\text{step}} = 40$, $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re } \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im } \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	40	p01	fallback-disabled	0.01565989	0.02133876	0.005914608	0.01869007	694	693.23
0.30	40	p01	fallback-enabled	0.008459005	0.02006063	0.01810559	0.01788859	0	684.56
0.30	40	p02	fallback-disabled	0.005432463	0.02378896	0.005517511	0.01778098	682	687.07
0.30	40	p02	fallback-enabled	-0.005268474	0.02161072	0.009747768	0.0167599	0	689.04
0.30	40	p03	fallback-disabled	0.008003131	0.02237772	-0.001666649	0.01764089	706	692.96
0.30	40	p03	fallback-enabled	-0.0004949058	0.02311059	-0.006990624	0.01735182	0	685.55
0.30	40	p04	fallback-disabled	0.008745843	0.02259568	-0.004489596	0.01845056	674	689.2
0.30	40	p04	fallback-enabled	0.003644535	0.02181327	-0.008153823	0.01734782	0	683.68
0.30	40	p05	fallback-disabled	-0.01903154	0.02017111	0.01317747	0.01699544	706	694.62
0.30	40	p05	fallback-enabled	-0.03129789	0.02037801	0.0119514	0.01630627	0	687.22
0.30	40	p06	fallback-disabled	-0.0002986763	0.02180983	0.02607715	0.01733579	748	692.13
0.30	40	p06	fallback-enabled	-0.007710652	0.02126615	0.01856183	0.01654142	0	688.96
0.30	40	p07	fallback-disabled	0.02532584	0.02124395	-0.02505056	0.01886911	762	689.74
0.30	40	p07	fallback-enabled	0.002687575	0.02085818	-0.01614941	0.01605725	0	690.56
0.30	40	p08	fallback-disabled	-0.008734219	0.02098861	-0.01013855	0.01874181	697	694.33
0.30	40	p08	fallback-enabled	-0.02535834	0.02008558	-0.005341889	0.01708532	0	690.5
0.30	40	p09	fallback-disabled	0.01068561	0.02173067	-0.0006874511	0.01730445	763	694.03
0.30	40	p09	fallback-enabled	0.01034041	0.02121209	0.007422074	0.01709937	0	682.42
0.30	40	p10	fallback-disabled	0.01235614	0.0233638	0.02023679	0.01809057	687	693.44
0.30	40	p10	fallback-enabled	0.005477779	0.02067539	0.02231379	0.01717797	0	690.72
0.30	40	p11	fallback-disabled	0.04796655	0.02126875	0.01827114	0.01868235	713	593.28
0.30	40	p12	fallback-disabled	0.0269696	0.02707341	0.00883942	0.01851616	703	580.3
0.30	40	p13	fallback-disabled	-0.03292448	0.0211817	-0.002185427	0.01689108	661	589.04

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.30	40	p14	fallback-disabled	0.003368151	0.02187849	-0.0166524	0.01759971	707	581.68
0.30	40	p15	fallback-disabled	0.008168649	0.02389712	0.0255379	0.02127426	714	582.84
0.30	40	p16	fallback-disabled	0.01921329	0.02193829	-0.006799498	0.01791733	749	586.7
0.30	40	p17	fallback-disabled	0.00324099	0.02351383	-0.02959712	0.02041043	659	586.47
0.30	40	p18	fallback-disabled	-0.01893928	0.02198779	0.003521362	0.01813265	708	586.95
0.30	40	p19	fallback-disabled	0.006099028	0.02173887	-0.01341522	0.02044137	761	589.91
0.30	40	p20	fallback-disabled	0.004389026	0.02101035	0.008088481	0.01767318	677	592.09

c. $\tau = 0.35$

TABLE X: Complete seed-level raw data at $\tau = 0.35$, $L = 2$,
 $N_{\text{chain}} = 50000$.

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.35	20	p01	fallback-disabled	1.10531	0.2318822	-0.04670139	0.108259	88818	658.72
0.35	20	p01	fallback-enabled	-0.1388314	0.01665358	-0.006483828	0.01116226	3375	1347.64
0.35	20	p02	fallback-disabled	1.188617	0.2487975	0.09598072	0.1097521	93419	651.41
0.35	20	p02	fallback-enabled	-0.1154849	0.01708691	-0.01198909	0.01244417	3383	1336.29
0.35	20	p03	fallback-disabled	1.072909	0.2241195	-0.1089058	0.09652858	73364	622.8
0.35	20	p03	fallback-enabled	-0.122215	0.02173926	-0.02725818	0.01060945	3442	1373.66
0.35	20	p04	fallback-disabled	1.255538	0.1892639	-0.2535048	0.1027938	98572	663.41
0.35	20	p04	fallback-enabled	-0.121592	0.01683719	0.007609468	0.01048514	3381	1326.89
0.35	20	p05	fallback-disabled	1.301381	0.2235782	0.2786242	0.1211668	91719	650.85
0.35	20	p05	fallback-enabled	-0.151568	0.01682003	-0.008203003	0.01298747	3209	1311.53
0.35	20	p06	fallback-disabled	1.410106	0.1642309	-0.01704563	0.133067	116076	695.38
0.35	20	p06	fallback-enabled	-0.1042772	0.01848972	-0.003100768	0.01076255	3382	1317.7
0.35	20	p07	fallback-disabled	0.9045214	0.2074573	0.04063163	0.06372107	82246	629.02
0.35	20	p07	fallback-enabled	-0.1090167	0.01801235	-0.006949602	0.01048871	3516	1354.95
0.35	20	p08	fallback-disabled	0.9654213	0.2484733	-0.1415433	0.07265464	70573	665.75
0.35	20	p08	fallback-enabled	-0.08576071	0.01943611	-0.01103185	0.01079167	3445	1352.88
0.35	20	p09	fallback-disabled	1.146976	0.2198932	-0.03298089	0.1131963	88812	693.94

τ	n_{step}	s	Setup	$\text{Re} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Re}	$\text{Im} \langle \mathcal{O}_{\text{SD}} \rangle$	σ_{Im}	F	T
0.35	20	p09	fallback-enabled	-0.1139017	0.01679789	0.0143521	0.01012423	3340	1303.45
0.35	20	p10	fallback-disabled	1.21394	0.2360242	-0.04741556	0.1094557	95687	678.16
0.35	20	p10	fallback-enabled	-0.1361935	0.0199941	0.005860844	0.0109517	3446	1313.73

Appendix B: Implemented Projection Policy in the Numerical Comparison

This appendix summarizes the projection policy used in the numerical comparison reported in Sec. IV. Its purpose is to make the implemented solver-side realization of the projection step explicit, while keeping the description at the level relevant for reproducing the reported results. Throughout this appendix, the local projection problem itself is the same as in Secs. II and III. What is specified here is only the implemented numerical policy used to solve the corresponding parametrized systems.

1. Position of the projection solve inside the molecular-dynamics update

At each molecular-dynamics substep, the integrator first advances to a trial configuration and then solves the projection constraint to return the configuration to the target manifold. If the projection solve succeeds, the substep continues. If it fails, the substep is terminated and the trajectory is treated as unsuccessful.

In the fallback-disabled setup, the projection step is attempted only in the standard parametrization. In the fallback-enabled setup, the same standard solve is attempted first, and an additional rescue policy is activated only after that primary solve fails. The numerical comparison in Sec. IV therefore distinguishes two implemented computational setups for the same local projection problem: one in which only the primary standard-side solve is used, and one in which that same primary solve is supplemented by a bounded rescue path.

2. Primary solve in the standard parametrization

The primary solve is performed in the standard parametrization by a simplified Newton iteration. The target tolerance used throughout the projection solve is

$$\text{tol} = 10^{-13},$$

and the initial iteration cap is

$$N_{\text{iter}}^{\text{std}} = 100.$$

Let r_k denote the residual norm at iteration k . The simplified Newton solve is declared successful when

$$r_k < \text{tol}.$$

The strict inequality is part of the implemented success criterion.

The iteration cap can be extended when the solve enters a near-solution regime. More precisely, once the residual satisfies

$$r_k \leq \max(200 \text{ tol}, 10^{-11}) = 10^{-11},$$

the allowed iteration count is increased in increments of 25, up to the hard ceiling

$$N_{\text{iter}}^{\text{std,max}} = 200.$$

Failure of the simplified Newton solve is determined by a combination of numerical gates rather than by a single condition. The solve is terminated unsuccessfully if any of the following occurs:

1. the projected linear step cannot be computed;
2. the flow evaluation fails;
3. the residual becomes non-finite;
4. repeated divergence is detected;
5. prolonged stagnation is detected together with tiny-step behavior;
6. the active iteration cap is reached without satisfying the convergence criterion.

The divergence gate distinguishes moderate and severe residual growth. With r_{k-1} and r_k denoting successive residual norms, the corresponding counters are updated when

$$r_k > 1.20 r_{k-1}$$

and

$$r_k > 1.50 r_{k-1},$$

respectively, with the severe-divergence condition additionally requiring

$$r_k > \max(10^5 \text{ tol}, 10^{-6}) = 10^{-6}.$$

The simplified Newton solve is terminated if the severe-divergence counter reaches 4 or the divergence counter reaches 10.

The stagnation gate uses the relative improvement

$$\frac{|r_{k-1} - r_k|}{\max(1, r_{k-1})}.$$

A stagnation counter is incremented when this quantity is smaller than 5×10^{-4} . A tiny-step counter is incremented when the step norm becomes smaller than

$$10\sqrt{\epsilon} \max(1, \|u\|_2),$$

where ϵ is the machine epsilon. The simplified Newton solve is terminated by the stagnation gate when all of the following are satisfied:

$$k \geq 30, \tag{B1}$$

$$N_{\text{stag}} \geq 24, \tag{B2}$$

$$N_{\text{tiny}} \geq 12, \tag{B3}$$

$$r_k > \max(10^4 \text{ tol}, 10^{-8}) = 10^{-8}, \tag{B4}$$

$$r_{\text{best}} > \max(20 \text{ tol}, 5 \times 10^{-11}) = 5 \times 10^{-11}. \tag{B5}$$

3. Implemented rescue policy after primary failure

In the fallback-enabled setup, an additional rescue policy is activated only after the primary simplified Newton solve has failed. This is the only rescue policy used in the numerical comparison.

The implemented rescue path has three parts.

First, an initial quasi solve is attempted with a limited iteration budget,

$$N_{\text{probe}} = 28.$$

Second, if the failed primary attempt already lies in the near-solution band, one additional full quasi solve is attempted with iteration budget

$$N_{\text{full}} = 100.$$

Third, if the failed primary attempt does not lie in that near-solution band, the rescue path may instead use one reduced-cost quasi solve with iteration budget

$$N_{\text{reduced}} = 36.$$

The fallback-enabled setup therefore uses a bounded rescue policy: after failure of the primary standard-side solve, the algorithm does not launch an unrestricted sequence of alternative nonlinear solves. Rather, it applies one probe attempt and then, depending on the position of the failed state relative to the near-solution band, at most one further quasi solve under a prescribed iteration budget.

Failure of the fallback-enabled setup therefore means that this rescue policy has given up under its prescribed gating and iteration-budget rules. It should not be interpreted as proof that no valid projected solution exists in principle.

4. Acceptance rule and stopping criteria in the quasi rescue solve

The quasi rescue solve uses the same target tolerance as the primary solve, but its acceptance rule differs slightly. A rescue attempt is certified as successful only if the residual satisfies

$$r \leq \text{tol},$$

together with successful reconstruction of the corresponding flowed state. Thus the rescue solve uses a non-strict residual test and also requires a successful recovery step.

Within each rescue attempt, the solver first evaluates the residual at the current candidate state. If this initial evaluation fails or the residual is non-finite, the attempt is terminated immediately.

Otherwise, the solver proceeds by a trust-region / Levenberg–Marquardt-type nonlinear least-squares update. A trial step is accepted if either of the following holds:

1. the trial residual already satisfies the target tolerance;
2. the actual reduction is positive and the reduction ratio is at least 10^{-2} .

If no acceptable step is found, the step size is reduced and the trial is repeated. The rescue attempt is terminated if the trust-region radius shrinks below

$$1.05 \times 10^{-6}$$

without obtaining an accepted step.

The rescue attempt is also terminated when stagnation persists under the local iteration budget. The stagnation limit is determined from the iteration budget N_{iter} of the current rescue attempt by

$$N_{\text{stag}}^{\text{quasi}} = \max\left(16, \min\left(40, \frac{N_{\text{iter}}}{2}\right)\right).$$

This gives

$$N_{\text{stag}}^{\text{quasi}} = 16 \quad \text{for the probe attempt,}$$

$$N_{\text{stag}}^{\text{quasi}} = 40 \quad \text{for the full rescue attempt,}$$

and

$$N_{\text{stag}}^{\text{quasi}} = 18 \quad \text{for the reduced-cost rescue attempt.}$$

After the local iteration loop terminates, the solver makes one final recovery attempt from the best state found during that run. The rescue attempt is declared successful only if this best residual satisfies the target tolerance and the corresponding flowed state can be reconstructed successfully. Otherwise the rescue attempt is declared unsuccessful.

5. Interpretive role of the implemented policy

The fallback-enabled and fallback-disabled setups differ through the solver-side policy applied after failure of the primary standard-side solve. In particular, the fallback-enabled setup supplements the primary simplified Newton solve by a bounded quasi-based rescue policy with explicit iteration budgets and acceptance gates. This is the implementation-level distinction used in the numerical comparison in Sec. IV.

-
- [1] M. Troyer and U.-J. Wiese, Phys. Rev. Lett. **94**, 170201 (2005), arXiv:cond-mat/0408370.
 - [2] P. de Forcrand, PoS **LAT2009**, 010 (2009), arXiv:1005.0539 [hep-lat].
 - [3] E. Y. Loh, J. E. Gubernatis, R. T. Scalettar, S. R. White, D. J. Scalapino, and R. L. Sugar, Phys. Rev. B **41**, 9301 (1990).
 - [4] A. Alexandru, G. Basar, P. F. Bedaque, S. Vartak, and N. C. Warrington, Phys. Rev. Lett. **117**, 081602 (2016), arXiv:1605.08040 [hep-lat].
 - [5] E. Witten, AMS/IP Stud. Adv. Math. **50**, 347 (2011), arXiv:1001.2933 [hep-th].

- [6] M. Cristoforetti, F. Di Renzo, and L. Scorzato (AuroraScience), Phys. Rev. D **86**, 074506 (2012), arXiv:1205.3996 [hep-lat].
- [7] Y. Tanizaki and T. Koike, Annals Phys. **351**, 250 (2014), arXiv:1406.2386 [math-ph].
- [8] A. Alexandru, G. Basar, P. F. Bedaque, G. W. Ridgway, and N. C. Warrington, JHEP **05**, 053, arXiv:1512.08764 [hep-lat].
- [9] M. Cristoforetti, F. Di Renzo, A. Mukherjee, and L. Scorzato, Phys. Rev. D **88**, 051501 (2013), arXiv:1303.7204 [hep-lat].
- [10] H. Fujii, D. Honda, M. Kato, Y. Kikukawa, S. Komatsu, and T. Sano, JHEP **10**, 147, arXiv:1309.4371 [hep-lat].
- [11] A. Alexandru, G. Basar, P. F. Bedaque, G. W. Ridgway, and N. C. Warrington, Phys. Rev. D **93**, 094514 (2016), arXiv:1604.00956 [hep-lat].
- [12] G. Fujisawa, J. Nishimura, K. Sakai, and A. Yosprakob, JHEP **04**, 179, arXiv:2112.10519 [hep-lat].
- [13] Y. Tanizaki, H. Nishimura, and J. J. M. Verbaarschot, JHEP **10**, 100, arXiv:1706.03822 [hep-lat].
- [14] M. Fukuma, PTEP **2024**, 053B02 (2024), arXiv:2311.10663 [hep-lat].
- [15] C. Cartis and L. Roberts, Mathematical Programming Computation **11**, 631 (2019).
- [16] C. Cartis, J. Fiala, B. Marteau, and L. Roberts, ACM Trans. Math. Softw. **45**, 10.1145/3338517 (2019).
- [17] M. Fukuma and N. Umeda, PTEP **2017**, 073B01 (2017), arXiv:1703.00861 [hep-lat].
- [18] M. Fukuma and N. Matsumoto, PTEP **2021**, 023B08 (2021), arXiv:2012.08468 [hep-lat].